

Infer.NET is a free and open source .NET software library for machine learning. It supports running Bayesian inference in graphical models and can also be used for probabilistic programming.

Overview

Infer.NET follows a model-based approach and is used to solve different kinds of machine learning problems including standard problems like classification, recommendation or clustering, customized solutions and domain-specific problems. The framework is used in various different domains such as bioinformatics, epidemiology, computer vision, and information retrieval.

Development of the framework was started by a team at Microsoft's research centre in Cambridge, UK in 2004. It was first released for academic use in 2008 and later open sourced in 2018. In 2013, Microsoft was awarded the USPTO's Patents for Humanity Award in Information Technology category for Infer.NET and the work in advanced machine learning techniques.

Infer.NET is used internally at Microsoft as the machine learning engine in some of their products such as Office, Azure, and Xbox.

The source code is licensed under MIT License and available on GitHub. It is also available as NuGet package.

See also

Comparison of machine learning software

Machine learning

ML.NET

scikit-learn

References

Further reading

Knowles, D.; Parts, L.; Glass, D.; Winn, John (2010). "Modeling skin and ageing phenotypes using latent variable models in Infer.NET" (PDF). Microsoft.

Winn, John; Minka, Tom (2009). "Probabilistic Programming with Infer.NET".

Winn, John; Simpson, Angela; Custovic, Adnan; Y. F. Tan, Vincent (2008). "Immune System Modeling with Infer.NET" (PDF). Microsoft.

External links

Infer.NET

GitHub - dotnet/infer

Machine Intelligence and Perception - Microsoft Research

Infer.NET - Practical Implementation Issues and a Comparison of Approximation Techniques